

# Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

## Supplementary Material

Anonymous CVPR submission

Paper ID \*\*\*\*\*

This document holds the material referred to from the main paper: the errors-in-variables treatment of the velocity regressor, the protocol behind the ablation measurement, the data paths of the DSEC measurements, and additional implementation and control analyses. The RVT source revision is b80f5683a6e2d5de65d4bde8105d796ccb50dbb1, and the SSM-ViT source revision is 7c871b55a0c5f00673c2c3975f6b6a1c5adbab88. Section numbers below refer to this document. References labeled main Sec.  $n$  refer to the main paper.

### 1. The velocity regressor

The regression of main Eq. 2 uses the finite-differenced label velocity  $\tilde{v} = v + \eta$ , where  $v$  is latent true velocity and  $\eta$  is label noise. Three error mechanisms enter, and they behave differently. The shared-noise term determines the estimator choice.

Classical dilution. Let the label center carry noise of standard deviation  $\sigma_c$  per frame and let the speed be formed by a central difference over  $\pm\Delta$ . The differenced speed then carries noise with variance

$$\sigma_\eta^2 = \frac{2\sigma_c^2}{(\Delta_f + \Delta_b)^2}, \quad (1)$$

and  $\tau$  is attenuated toward zero by  $\lambda_{\text{att}} = \text{Var}(v)/(\text{Var}(v) + \sigma_\eta^2)$ . The noise is the label noise of the dataset the fit runs on, so it is measured there: Sec. 3 solves for  $\sigma_c = 0.6515\text{px}$  on Gen1 itself, and Eq. 1 is evaluated row by row rather than assuming a fixed cadence. On the reported rows  $\text{Var}(\eta)$  is 1.0 against a  $\text{Var}(\tilde{v})$  of 110.6 in  $\text{px}^2/\text{s}^2$ , so  $\lambda_{\text{att}} = 0.991$ . Because the reported Eq. 2 also includes geometric controls and per-sequence effects, this marginal calculation is not the exact attenuation factor for the final multivariable coefficient. We use it only as a scale diagnostic showing

that label-velocity noise is small relative to marginal velocity variation. Under the synthetic setup of Sec. 4, the centered estimator recovers 0.92 of the injected lag against the predicted 0.933.

Smoothing bias. On the uniform grid used for the label-noise calibration, a central difference returns the mean velocity over  $2\Delta$  rather than the instantaneous velocity. Expanding the trajectory about the query instant,

$$\frac{x(t + \Delta) - x(t - \Delta)}{2\Delta} = \dot{x}(t) + \frac{\Delta^2}{6} \ddot{x}(t) + O(\Delta^4), \quad (2)$$

so the regressor carries a systematic term of order  $\Delta^2 \ddot{x}/6$ . That is a bias in the regressor, not noise in it, and no errors-in-variables correction removes it. It is made small by making  $\Delta$  small, which Eq. 1 says makes the dilution worse. The reported regression rows use the local secant  $[x(t + \Delta_f) - x(t - \Delta_b)]/(\Delta_f + \Delta_b)$ . Their median  $|\Delta_f - \Delta_b|$  is zero, so the first-order acceleration term is absent in the typical row. The general expansion includes  $(\Delta_f - \Delta_b)\ddot{x}/2$  and is covered by the stated local-secant interpretation.

The shared-noise term. The first two mechanisms are properties of the regressor alone. The third couples the regressor to the response. A forward difference  $(c_{k+1}^* - c_k^*)/\Delta$  carries the label noise  $\delta_k$  of the current center with weight  $-1/\Delta$ , and the residual  $\hat{c} - c_k^*$  carries the same  $\delta_k$  with weight  $-1$ , so

$$\text{Cov}(\hat{c} - c_k^*, v_f) = \frac{\text{Var}(\delta)}{\Delta} \quad (3)$$

even at  $\tau = 0$ , and the induced slope is  $\text{Var}(\delta)/\Delta \text{Var}(v)$ . A backward difference carries  $\delta_k$  with the opposite sign. The centered difference omits  $c_k^*$  and removes the direct same-frame shared-noise

term. Residual bias from temporally correlated annotation errors is not identified by this construction, which is why the forward and backward fits are reported beside it as a control.

Instrumental-variable alternative. An instrumented estimate would replace the current speed by a velocity finite differenced from a strictly disjoint earlier label pair, on the argument that disjointness implies independent errors. DSEC provides a counterexample to assuming that disjoint label errors are generally independent. The third-difference kernel  $(1, -3, 3, -1)$  forces  $\rho(1) = -0.75$  for white noise, the same code returns  $-0.740$  on white noise and  $-0.3566$  on the DSEC-Det labels, and pooled-ratio bias, heteroscedastic segments and heavy tails each returned the white-noise value through the identical pooling code, so the departure is a property of the labels. That measurement does not identify what the correlation structure is, beyond excluding a first-order autoregressive form. The paper does not report an instrumental estimate.

The construction term. A benchmark whose high-rate labels are constructed between anchors adds a term to  $e_{||}$ , where  $e_{||} = \langle \hat{c} - c^*, \hat{u} \rangle$ , that has nothing to do with the predictor. The chord of a curved image trajectory runs ahead of the truth throughout the interval and meets it at the anchors, which produces a signed, frame-locked, resetting residual for a predictor with no lag. Main Sec. 4.1 shows that exact adjacent-frame linear interpolation is inconsistent with the observed DSEC-Det label trajectories as the sole construction mechanism, and the Gen1 measurements of main Sec. 3 are taken at label times.

## 2. The influence profile

Main Sec. 3.3 measures the window-ablation sensitivity profile of the released rvt-t checkpoint. The stored input is a stacked histogram of ten bins of 5 ms covering the 50 ms preceding the query time. For each sample the recurrent state is advanced over eight preceding steps so that the network is not read out of a zero state, the reference output is recorded, and then each bin in turn is zeroed and the output recorded again. A bin's influence is the magnitude of the change its ablation produces in the emitted detection tensor, relative to its norm, averaged over 480 samples drawn from the first twelve lexicographically ordered released Gen1 validation sequences, using the first forty post-warm-up samples from each sequence. This deterministic rule gives 480 samples. The measured ablation-sensitivity centroid is the weighted average  $\sum_k s_k c_k / \sum_k s_k$  over

the ten bin centers  $c_k$ , where  $s_k$  denotes the measured non-negative bin-ablation sensitivity. The ablated pass starts from the recurrent state entering the step, not from the state the reference pass returned. The two passes therefore share one history and differ only in the ablated bin. If the returned state is used instead, the passes are one step out of phase and the centroid moves by  $+1.19$  ms.

For the recurrent-history measurement, the same deterministic sampling rule uses 576 samples, the first forty-eight post-warm-up samples from each of the same twelve validation sequences. The current window and the preceding nine windows form the 500 ms horizon, and the current window and preceding nineteen windows form the 1000 ms horizon. For each lag, the corresponding whole input window is zeroed while the saved reference history before that lag is restored, and the state is propagated through the intervening windows to the target step. The relative  $L_2$  change in the target detection tensor is averaged over samples to give a non-negative window sensitivity  $q_\ell$ . The horizon centroid is

$$c_{q,H} = \frac{\sum_{\ell=0}^{L_H-1} q_\ell [-(25 + 50\ell) \text{ ms}]}{\sum_{\ell=0}^{L_H-1} q_\ell},$$

where  $L_H$  is the number of windows in horizon  $H$ , and the newest-window share is  $q_0 / \sum_\ell q_\ell$ . The same saved reference history is used for the reference and ablated passes at each lag.

Two fills define  $q_\ell$ , and the manuscript reports both. Under the zero fill the selected window is set to zero. Under the swap fill it is replaced by the window at the same within-chunk index drawn from an unrelated sample in a different validation sequence, which preserves a real window's bin statistics and dynamic range while removing that window's own content. Everything else is held identical across the two: the same lag, the same restored reference history, the same propagation through the intervening windows, and the same relative  $L_2$  read-out. The two are interventions rather than ground truth and they do not agree on scale, which is why both are reported. From lag 0 to lag 19 the zero fill decays by a factor of 10.3 and the swap fill by 2.2, and the ratio of swap to zero sensitivity rises from 2.43 at lag 0 to 11.47 at lag 19. Under the swap fill a window a second earlier still carries 46 % of the newest window's sensitivity. Substantial long-lag sensitivity therefore persists under the on-manifold intervention, although its absolute magnitude remains instrument-dependent.

Two properties of that measurement bound it. Ablation reports the sensitivity of the output to a bin. The ablation profile is an operational input-sensitivity mea-

sure and is not, in general, a decomposition of temporal weights. For a nonlinear predictor it is a local rather than a global description. And a single-bin ablation leaves the remaining bins in place, so the measurement is a partial derivative rather than a decomposition of the output.

Because the profile is defined by an intervention, the centroid is reported under four instruments rather than one. Zeroing a bin gives  $-23.81$  ms, setting it to its dataset mean  $-26.35$  ms, replacing it with the same bin of an unrelated sample  $-25.01$  ms, and the gradient of the output norm with respect to the bin, which ablates nothing at all,  $-22.35$  ms. The four span  $4.01$  ms. Two of them remove content, one substitutes foreign content and one perturbs nothing, so their agreement is not a property shared by construction. The property the manuscript uses the centroid for, that it sits tens of milliseconds before the label instant and well away from the fitted temporal coefficient, holds under all four.

The sampling rule is deterministic rather than random, so the sequences are the resampling unit for its uncertainty. Resampling the 12 sequences with replacement gives a 95 % interval of  $[-24.07, -23.56]$  ms for the zero-fill centroid, a standard error of  $0.13$  ms, and no interval wider than  $0.51$  ms under any of the four instruments. Dropping any one sequence moves the estimate by at most  $0.08$  ms, and the individual sequence centroids run from  $-24.63$  to  $-23.20$  ms. The choice of instrument therefore moves this centroid by  $4.01$  ms while the choice of sequences moves it by well under one. The instrument spread is the binding uncertainty rather than the sampling error. Sequences outside the twelve are a separate question, answered only by measuring them.

Masking the leading bins of the stored input is not a route to the same quantity at a shorter support. For a fixed network of influence weights  $a_k$  on bin centers  $c_k$ , let a constant-velocity trajectory be  $p(t) = p_0 + vt$ . The conditional expectation after masking the leading  $j$  bins is  $A_j p_0 + C_j v$  with  $A_j = \sum_S a_k$  and  $C_j = \sum_S a_k c_k$  over the retained set  $S$ . Reading a time off that requires the normalization  $C_j/A_j$ , which the network need not perform, and the result equals the retained-mass centroid only for non-negative weights.

### 3. The label-center noise of Gen1

The errors-in-variables treatment of Sec. 1 and the calibration of the velocity estimator both need the dispersion of the Gen1 label center about a smooth trajectory. The value solved for elsewhere in this work,  $0.4974$  px, is DSEC-Det's, and does not transfer to a different annotation effort on a different sensor, so it is solved again on Gen1.

For a track center series  $x_i = f(t_i) + e_i$  on a uniform grid, the  $k$ -th difference has  $\text{Var}(D_k) = \binom{2k}{k} \sigma^2$  plus a motion term of order  $k$ , so  $\text{Var}(D_k)/\binom{2k}{k}$  falls toward  $\sigma^2$  from above. Halving the sampling rate multiplies the motion term by  $4^k$  and leaves  $\sigma^2$  untouched, giving a second equation in the same two unknowns,  $\sigma^2 = (4^k V_{\text{full}} - V_{\text{half}})/(4^k - 1)$ . Tracks are rebuilt by the same-class overlap linking the output-time measurement uses, giving 691 tracks of at least twelve labels on uniform annotation grids with cadences of 250, 500, or 1000 ms, corresponding to 4, 2, or 1 Hz. The reported regression rows instead use the two adjacent label neighbors with variable intervals. Their row-wise noise variance is  $2\sigma_c^2/(\Delta_f + \Delta_b)^2$ , rather than a fixed-cadence approximation. Variances are trimmed at 5 %, because all four Gen1 box coordinates are stored as integers and a median absolute deviation on lattice-valued data locks onto discrete values.

The two-rate elimination agrees with the full-rate ladder to four decimals from  $k = 5$  onward, which is the check that the motion term is negligible rather than assumed away. The ladder extrapolates to  $\sigma = 0.6515$  px, of which  $0.1443$  px is the quantization of a  $0.5$  px center lattice, leaving  $0.6353$  px of annotation dispersion.

### 4. The sign-flip calibration

Equation 3 predicts an artifact whose size is set by the label noise and the observed velocity variance, so it is calibrated before the estimator is used. 2000 synthetic tracks of fourteen labels on the Gen1 grid are generated with quadratic motion. The synthetic experiment calibrates the shared-noise artifact and attenuation under quadratic motion. The simulated detector reports the true position at  $t - \tau_{\text{true}}$  with its own independent error, and the label centers are corrupted at  $0.6515$  px.

With  $\tau_{\text{true}} = 0$  the forward estimator returns  $-5.64$  ms and the backward one  $+5.84$  ms, summing to  $+0.20$  ms, against the closed form  $-5.63$  ms, the centered estimator returns  $+0.00$  ms. Setting the label noise to zero returns all three to  $+0.02$  ms, and the placebo stays at its null throughout, so it does not absorb the artifact and remains usable on real rows. In the synthetic physical-lag experiment, the centered estimator recovers  $0.92$  of the injected lag, consistent with the predicted attenuation of  $0.933$  for this noise.

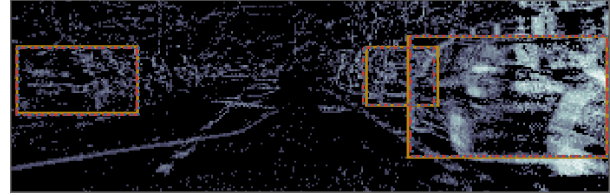
### 5. The output-time specification sweeps

Lag recovery under the full control set. Adding  $-\tau_{\text{inj}} \tilde{v}$  to the observed residual creates a known perturbation along the fitted temporal regressor. The coefficient is recovered at  $1.000$  of its injected value at 10, 25, 50 and 100 ms, showing that the geometric controls do

not absorb this regressor. Injecting instead a purely geometric displacement of 0.05 px per px of box height moves the minimal specification's  $\tau$  by +0.63 ms and the reported one's by less than 0.01 ms.

Sensitivity to axis pairing. The reported specification pairs box height and image row with  $\hat{y}$  and box width and image column with  $\hat{x}$ . Entering all four on both axes gives  $-3.62 \pm 7.41$  ms. Dropping each in turn moves  $\tau$  by under 0.3 ms except for the image column, whose removal returns the placebo to its inflated value. Raising the overlap floor to 0.7 gives  $+0.49 \pm 4.67$  ms, the tightest of the sweep. Across every specification carrying geometric controls, including speed floors from 5 to 20 px/s,  $\tau$  spans  $-20.2$  to  $+11.1$  ms, and none of them reaches the ablation-sensitivity centroid.

Association independent of nominal-time overlap. The rows are built by matching a detection to a label at the label's nominal time, so a gate on overlap could in principle discard exactly those detections whose temporal offset is large and select  $\tau$  toward zero. Raising the floor cannot detect that, because it discards more. The informative direction is to open the gate or to remove it. `src/e62_association.py` rebuilds the rows from the five-checkpoint detection dump of Sec. 11 under 14 association rules and refits the reported specification under each: greedy and Hungarian assignment at overlap floors from 0.05 to 0.50, and a rule that never tests overlap at all, admitting a detection whose center falls within a fixed multiple of the label's box scale. That dump is re-thresholded to the regression's confidence floor of 0.1, but its non-maximum suppression was run at 0.65 against the 0.45 of the dump behind main Tab. 1, and re-thresholding does not undo the looser suppression. The rebuild's own 0.30 gate therefore keeps 22 518 rows against the 22 534 of the reported table and returns  $-2.96 \pm 7.12$  ms against  $-2.40 \pm 7.18$  ms, a difference of 0.08 standard errors. The sweep is read within the rebuild, as a comparison across association rules on one detection set. Across all 14 rules  $\tau$  spans  $-8.96$  to  $-0.90$  ms, a spread of 8.06 ms against a median standard error of 7.12 ms. Every rule leaves  $\tau$  within two standard errors of zero and leaves the ablation-sensitivity centroid more than two standard errors away. The overlap-free rule admits 22 895 rows against 22 518 for the rebuild's own gate and gives  $-5.58 \pm 7.86$  ms against  $-2.96 \pm 7.12$  ms, a difference of 0.37 standard errors. Under the minimal specification the same sweep spans 12.01 to 16.35 ms, so neither specification's conclusion is an artifact of the matching rule. Fig. 1 draws one frame of the sweep.



released detections  
ground truth at the label instant  
ground truth at the newest window's centroid,  $\delta = -23.81$  ms

Figure 1. One validation frame of the sweep of Main Fig. 1b: released detections (solid), ground truth at its label instant (dashed) and displaced to the newest window's ablation-sensitivity centroid (dotted). The frame is chosen by rule, the largest median label speed among frames with two labels above 20 px/s, its fastest label moves 45.2 px/s, so the two placements sit 1.08 px apart.

The bootstrap. Resampling sequences with replacement, 4000 times, gives a 95% interval of  $[-17.0, +11.5]$  ms with 3 draws reaching the influence centroid. The fit absorbs the per-sequence constant vectors by within-cell demeaning, which reproduces the full design's coefficient to 0.007 ms.

## 6. The per-object ratio

For comparison with the estimator in main Eq. 2, we evaluate the per-object ratio  $-(\hat{\mathbf{c}} - \mathbf{c}^*, \hat{\mathbf{u}}) / \|\hat{\mathbf{v}}\|$  on the same 22 534 rows. Across these rows, it has a sample standard deviation of 4170 ms and an interquartile range of 430 ms. Its mean over all rows is  $+24.22 \pm 27.73$  ms, close to the  $+23.81$  ms coefficient corresponding to the newest-window ablation-sensitivity centroid under the convention of main Eq. 2. Above a 15 px/s floor it gives  $+13.34 \pm 6.17$  ms. Because the ratio has no intercept, a velocity-independent spatial bias can appear as an apparent temporal offset after division by speed. Main Eq. 2 includes an intercept.

## 7. The port of the released checkpoint

The released rvt-t checkpoint loads into a model built from the repository's own configuration files with 0 missing and 0 unexpected keys at 4.41 M parameters, and runs forward under a current framework without the training code it was serialized from. Both predictor measurements run on that port. A key count is a structural check, so two numerical ones are reported alongside it.

Parameter counts against the published ones. Summing the tensors each checkpoint carries gives 4.41, 9.87 and 18.54 M for the three RVT releases and 9.68 and 18.19 M for the two SSM-ViT releases. The largest deviation from the counts printed in the two papers is

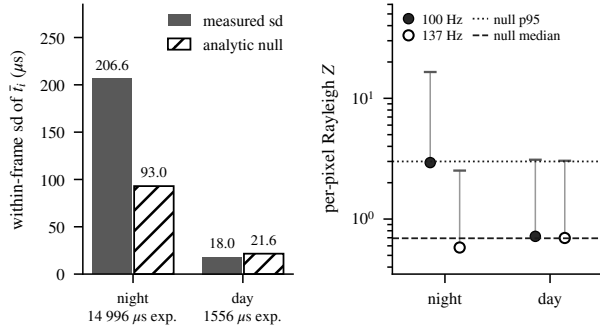


Figure 2. Left: within-frame dispersion of  $\bar{t}_i$  against its analytic null in the two exposure strata, 1134 frames of `zurich_city_09_a` at 14996  $\mu$ s and 20 frames of `interlaken_00_c` at 1556  $\mu$ s. The daytime frames are selected by retaining frames with at least three qualifying boxes. Right: per-pixel Rayleigh statistic  $Z = R^2/N$  at 100 Hz and at 137 Hz for the same two sequences, median with a whisker to the ninety-fifth percentile, against the Exp(1) null. Both contrasts run in the same direction, and Main Sec. 4.4 reports the controls applied to that dispersion.

0.051 M, so the ported models are the architectures the releases describe rather than approximations of them.

Independent evaluator cross-check. Every level in this paper comes from its own evaluator, so each detection dump is rescored with `pycocotools` at  $AP@[.50:.95]$  on the identical arrays. For `rvt-t` the two give 38.95 and 38.90 points, and across all five checkpoints they disagree by at most 0.101 points. The reported levels are therefore a property of the detections, not of the scorer.

Validation-split and checkpoint-port limitations. The published Gen1 tables are reported on the test split. The release distribution available to this work carries the validation split alone, 429 sequences with no test directory, so the levels above are validation-split levels and the published numbers are used only for the parameter comparison and for the ordering within RVT, which is preserved. A pass of the original training-code path on identical samples, to give a per-sample output difference against the port, requires the training framework the checkpoints were serialized from and is left to the artifact release. Of the 429 released validation sequences, 406 enter the stored-detection evaluation used for sequence resampling. The remaining 23 are excluded because they contribute no stored-detection rows. All bootstrap results in Secs. 12–13 resample these 406 sequences.

## 8. The DSEC measurements

Data path. For each labeled frame of a sequence, the frame’s own exposure interval  $[a, b]$  is read from the

| Arm                                           | frames | sd    | null  | excess |
|-----------------------------------------------|--------|-------|-------|--------|
| Ceiling sequence, all pixels                  | 1134   | 206.6 | 93.0  | 183.6  |
| Phase-locked pixels removed                   | 1134   | 205.1 | 97.1  | 180.6  |
| Random pixels, matched size                   | 1133   | 206.3 | 94.8  | 182.9  |
| Random-position boxes                         | 1123   | 338.2 | 134.5 | 306.3  |
| by each box’s own 100 Hz modulation depth $m$ |        |       |       |        |
| $m \in [0.122, 0.410)$                        | 270    | 119.3 | 74.6  | 93.1   |
| $m \in [0.410, 0.467)$                        | 243    | 149.1 | 76.2  | 128.1  |
| $m \in [0.467, 0.534)$                        | 207    | 156.6 | 85.1  | 131.5  |
| $m \in [0.534, 1.050]$                        | 265    | 227.2 | 91.5  | 208.0  |

Table 1. Per-object event centroid inside one published exposure window of `zurich_city_09_a`, in  $\mu$ s, each entry the median over frames of a within-frame statistic against that frame’s own null  $w^2/12N_i$ . The first block’s excess follows the main Sec. 4.4 statistic summarized over frames. Rows two and three remove the 14334 pixels phase-locked at 100 Hz and a random subset of that size, row four places boxes of the same shapes at random positions. In the lower block a frame enters a stratum when 3 of its boxes fall in it, so those frame counts do not partition the first row’s.

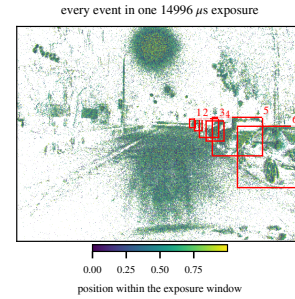


Figure 3. Every event inside one published 14996  $\mu$ s exposure window of `zurich_city_09_a`, colored by its position within that window, with that frame’s released boxes numbered left to right. The frame is the median-dispersion one of 602 eligible candidate frames with at least six qualifying boxes, not the largest.

| Sequence                      | boxes | $m$   | $Z_{100}$ | $Z_{137}$ | locked   |
|-------------------------------|-------|-------|-----------|-----------|----------|
| <code>zurich_city_00_a</code> | 2914  | 0.428 | 117.6     | 3.15      | 100.00 % |
| <code>zurich_city_01_a</code> | 2134  | 0.440 | 99.1      | 4.69      | 99.63 %  |
| <code>zurich_city_02_a</code> | 619   | 0.445 | 68.4      | 2.75      | 99.68 %  |
| <code>zurich_city_03_a</code> | 241   | 0.489 | 310.9     | 25.85     | 95.44 %  |
| <code>zurich_city_09_a</code> | 6997  | 0.466 | 251.5     | 7.82      | 99.23 %  |
| <code>zurich_city_10_a</code> | 3611  | 0.438 | 111.6     | 3.33      | 100.00 % |

Table 2. Every DSEC training sequence whose exposure is pinned at 14996  $\mu$ s, measured per labeled box over the events inside that box during that frame’s own published exposure window, with at least 400 events per box.  $m$  is the median over boxes of the modulation depth  $m = 2|C|$  at 100 Hz.  $Z_{100}$  and  $Z_{137}$  are the median over boxes of the Rayleigh statistic at 100 Hz and at the off-frequency control on the same events. The analytic Exp(1) median of 0.693 does not describe either column, which is why the two are compared with each other. The locked fraction denotes the fraction of boxes whose local excess at 100 Hz exceeds the ninety-ninth percentile of the same quantity at 137 Hz.

dataset’s published `image_exposure_timestamps_left`

file, the event stream is sliced to that interval through the release's `ms_to_idx` index, and the events inside each released box are collected. The `ms_to_idx` index resolves to whole milliseconds, so the slice it returns is filtered to  $a \leq t < b$  in microseconds before any statistic is formed, and the analytic null is computed for the same interval. Events and frames share one clock in this dataset, so no calibration step enters. For the dispersion of main Sec. 4.4 a box contributes only if it carries at least 200 events in the window, and a frame only if at least 3 boxes qualify, the reported statistics are medians over frames of a within-frame quantity, so no frame is weighted by its object count. For the per-box statistics of Table 2 a box contributes with at least 400 events and no frame-level requirement applies. The selected qualitative frame is shown in Supplement Fig. 3, and the paired daytime control is shown in Supplement Fig. 2.

For frame  $f$ , let  $I_f$  be the qualifying boxes, let  $N_{fi}$  be the event count inside box  $i$ , and let  $\bar{t}_{fi} = N_{fi}^{-1} \sum_{k \in i} t_k$  be its event-time centroid relative to the frame's mid-exposure. The observed within-frame dispersion and its frame-specific analytic null are

$$d_f = \left[ \frac{1}{|I_f| - 1} \sum_{i \in I_f} (\bar{t}_{fi} - \bar{t}_f)^2 \right]^{1/2},$$

$$n_f = \left[ \frac{1}{|I_f|} \sum_{i \in I_f} \frac{w_f^2}{12N_{fi}} \right]^{1/2}.$$

where  $\bar{t}_f = |I_f|^{-1} \sum_{i \in I_f} \bar{t}_{fi}$  and  $w_f = b_f - a_f$ . The frame-level excess is  $e_f = [\max(d_f^2 - n_f^2, 0)]^{1/2}$ , and the reported entries are the medians of  $d_f$ ,  $n_f$ , and  $e_f$  over eligible frames.

The stereo exposures. The left and right exposures open at the same microsecond in 100% of the 7080 frames checked and differ only in when they close, and the published image timestamp equals the mean of the two mid-exposures to within 1  $\mu$ s.

Archive version and track association. DSEC-Det ships more than one annotation version. The numbers in main Sec. 4.1 describe the archive `dsec-det_left_object_detections.zip`, 4 655 023 bytes, retrieved 2026-09-01. Whether those labels are tracker output on the RGB frames, which the dataset's own description states, is not distinguishable from human annotation by a second-difference residual. On the Gen1 side the `track_id` field is a per-frame index rather than a persistent identifier, so association for the output-

time fit is rebuilt by same-class overlap between adjacent label times, at which 79.0% of boxes link.

Null. For events uniform on  $[a, b]$  the mean of  $N_i$  times has variance  $w^2/12N_i$  with  $w = b - a$ . Each frame therefore carries its own null, formed from the published width and the observed counts, and the excess of main Sec. 4.4 is summarized over frames rather than formed from the medians of the two summaries. That is why the excess column of Table 1 is not  $(sd^2 - null^2)^{1/2}$  of the two columns beside it.

Phase test. The per-pixel test at frequency  $f$  takes the event phases  $\phi_k = 2\pi f t_k \bmod 2\pi$  of a pixel, forms the resultant length  $R$  and reports  $Z = R^2/N$ , which is  $\text{Exp}(1)$  under uniform phase. Raw modulation depth is not usable at these counts, because for  $N$  events at random phase the resultant is already of the size of the effect. Pixels enter with at least 100 events over the tested slice, and every fourth event is used to reduce sample size. The off-frequency and daytime controls remain close to their nominal reference under the same procedure. Two arms hold the estimator to its null, an off-frequency arm on the same events and the same frequency on daytime events. The fraction of locked pixels is reported relative to live pixels in main Sec. 4.2 and separately relative to the full sensor. The two denominators differ, and each statement identifies its denominator.

The per-box statistics of Table 2. The same estimator is applied to the events inside one released box during one published exposure window rather than to the events of one pixel over a long slice. For box  $i$  the depth is  $m = 2|C|$  and the statistic is  $Z = N_i|C|^2$  with  $\omega = 2\pi f$  and  $C = N_i^{-1} \sum_{k \in i} e^{-i\omega t_k}$ , phases referred to that frame's mid-exposure instant. A finite exposure does not produce uniform phase at a tested frequency, so the pixel-level  $\text{Exp}(1)$  null is not assumed for these per-box values. The depth is reported descriptively, without a random-phase floor. Boxes are matched to exposure windows by nearest mid-exposure instant within half a frame period. The off-frequency arm at 137 Hz is run on exactly the same events. The per-box 137-Hz arm is used as a same-event descriptive control rather than as an independent spectral-resolution test. The illumination attribution is supported primarily by the long-slice 100-Hz harmonic structure and the daytime control.

Pixel exclusion and modulation strata. The exclusion mask of Table 1 is the set of pixels reaching  $p < 10^{-3}$  in the per-pixel test, and the matched control draws a

random pixel set of exactly the same size, so the two arms remove the same number of pixels. The mask is built from a six-second slice at every fourth event. The stratification uses no mask: the quartiles of  $m$  are taken over all 6841 qualifying boxes, and a frame contributes to a stratum when at least 3 of its boxes fall inside it, which is why the frame counts of that block are not a partition of the first row's.

## 9. The permutation null

The analytic null  $w^2/12N_i$  for the main Sec. 4.4 statistic is the variance of a mean of  $N_i$  draws uniform on the exposure, so it assumes the events of a box arrive independently and uniformly.

The permutation null does not assume independent or uniform event arrivals. Instead, it conditions on the pooled frame-level event times and each box's event count while randomizing box assignment. For each frame the events of all qualifying boxes are pooled and reassigned to boxes at random, preserving each box's count. The exposure, the flicker and the scene's burstiness all survive that reassignment, only which box an event belongs to is destroyed. Over 1134 frames and 200 draws each it returns  $78.6 \mu\text{s}$ , 0.85 times the analytic null rather than above it, and 94.1% of frames exceed their own permutation null. A burst common to the frame displaces every  $\bar{t}_i$  in the same direction and cancels in their dispersion, which is why the statistic was already insensitive to it. For a separate random-position draw evaluated against the permutation null, boxes of the same shapes give  $339.4 \mu\text{s}$  against  $113.3 \mu\text{s}$ . Thus the dispersion is not specific to annotated object regions, although object regions may still affect it.

## 10. The carried state of the SSM release

Main Sec. 3.3 measures five released checkpoints of two architectures. The second is SSM-ViT (Zubic et al., CVPR 2024), a fork of the RVT repository in which each stage's ConvLSTM is replaced by an S5 layer. It distributes RVT's own preprocessed Gen1 and builds its window boundaries with the same statement of the same script, so the window convention of main Sec. 3.2 holds there unchanged.

Its recurrent state does not. Let  $s$  denote the carried state entering the first step of a chunk. That release injects it as  $\bar{\Lambda}_0 \leftarrow \bar{\Lambda}_0 \odot s$  before an associative scan whose binary operator is  $(a_i, b_i) \circ (a_j, b_j) = (a_j a_i, a_j b_i + b_j)$ . The scan's outputs are the  $b$  components, and  $b_p$  is built from  $a_1, \dots, a_p$  and  $b_0, \dots, b_p$ ,  $\bar{\Lambda}_0$  enters only the  $a$  component of the first element and therefore no output. The carried state is inert.

We verify this code-level property empirically. Driv-

ing each model the way its own release drives it and zeroing the state entering a chunk of 8 windows, the relative  $L_2$  change in the emitted detection tensor is 0.00000 at every position for SSM-ViT, and 0.0616 at position one falling to 0.0304 at the last position for rvt-t. A model that ignored its history altogether would return zero to both arms, so the control ablates the window one position earlier inside the same chunk: SSM-ViT answers 0.0276 there, falling to 0.0106, so it is recurrent within a chunk and only across chunks is the state lost.

## 11. Effect of temporal displacement on checkpoint ordering

Main Sec. 3.5 reports that displacing the ground truth to the newest window's ablation-sensitivity centroid costs 0.04 points on one checkpoint. Whether the interval changes a comparison needs more than one detector, so all 5 released checkpoints of main Sec. 3.3 were run over the Gen1 validation split under one protocol: the same frames, the same confidence floor of 0.01 and non-maximum-suppression overlap of 0.65, and ground truth that is identical across the five and identical to the dump main Sec. 3.5 already uses. Each was then scored at the label instant and again at the displacement it prefers, which ranges from  $-15$  to  $0$  ms across the five.

| stratum        | ordering at the label instant |
|----------------|-------------------------------|
| all moving     | $b > s > t > S5-B > S5-S$     |
| 10–25 px/s     | $S5-B > b > t > S5-S > s$     |
| 25–50 px/s     | $S5-B > b > S5-S > s > t$     |
| $\geq 50$ px/s | $b > t > S5-S > s > S5-B$     |

Table 3. Ordering of the 5 released checkpoints at the label instant, by object speed,  $t$ ,  $s$ ,  $b$  are the RVT capacities and S5-S, S5-B the SSM ones. Scoring each at its own preferred displacement leaves every one of these orderings unchanged. A common displacement sweep reveals one localized inversion between rvt-b and S5-S in the 25–50 px/s stratum. Ranking varies across speed strata, and the common sweep produces no other pairwise inversion. s5vit-base is first in the middle two strata and last in the fastest, and rvt-t is third overall, last between 25 and 50 px/s and second above 50.

The own-preferred-offset scores preserve the validation-split ordering in each speed stratum (Tab. 3). Under a common displacement sweep, the 25–50 px/s stratum contains a localized ranking inversion between rvt-b and S5-S. The other evaluated strata contain no sign-changing pair. The aggregate ranking and the fast-object ranking have different leaders, so the strata are reported separately.

The released streaming evaluation steps through the split in non-overlapping chunks of 21 win-

dows (sequence\_length in config/dataset/gen1.yaml, start\_indices in sequence\_for\_streaming.py steps by exactly that). The evaluated input sensitivity of a released detection therefore depends on where its frame falls inside that chunk, from a full 21 windows at the last position to the current window alone at the first. This is reported as a property of the release that its documentation does not state, which is the same class of undeclared quantity the rest of this paper measures, and not as a claim about what the architecture could support if the state were carried.

## 12. Timing differences against the sampling uncertainty of the score

Main Sec. 3.5 reports that displacing the ground truth across the measured interval moves mAP by 0.04 points and reorders no checkpoint. On a fixed split both are deterministic functions of the stored detections, so neither is interpretable without the uncertainty that attaches to reading them as statements about the sequence population the split samples. Frames inside a sequence are dependent, so the resampling unit is the sequence: a cluster bootstrap over the 406 validation sequences was run for all 5 checkpoints at each displacement, resampling sequences with replacement and rescored from the stored detections.

Two standard errors must be distinguished. A claim made within one paper compares that paper's own score at two displacements on one set of detections, so the relevant quantity is the paired  $SE(\Delta mAP)$ , which is 0.035–0.058 points across the five checkpoints. A claim that compares across papers compares two absolute numbers, and the relevant quantity is  $SE(mAP)$ , which is 2.19–2.82 points, fifty times larger.

Two quantities that are easy to conflate are therefore reported separately, because they are referred to different baselines. The first is the displacement from the maximum of the sweep at which the loss reaches two standard errors: 13.7–19.8ms paired and 153–174ms in the single-score sense. The second is the effect of the measured interval itself, referred to zero displacement, which is 0.041–0.081 points across the 5 checkpoints against a paired two-standard-error band of 0.070–0.116 points. The second clears its band for 1 of 5 checkpoints. The measured interval therefore sits at the paired sampling-resolution floor under sequence resampling. The identifiability conclusion rests on where the evidence lies rather than on that comparison. The 0.04-point figure measures the sensitivity of the pooled benchmark to the interval, which follows from the benchmark's speed composition (main Tab. 2) and places no bound on the interval itself.

The same computation gives the timing difference

a leaderboard could resolve. Let  $D^*$  be the smallest common displacement at which an adjacent pair in the validation-split ordering exchanges places. Over all moving boxes  $D^* = 43.7\text{ms}$ ,  $1.83\times$  the measured interval, and in the 25–50 px/s stratum  $D^* = 11.2\text{ms}$ , or  $0.47\times$ . A benchmark whose ordering only responds to timing displacements of this size cannot be used to certify that a timing convention is immaterial.

That ordering is itself unstable. The reference ordering here is the one the 5 checkpoints take on this validation split at the nominal label time, not the ordering of the published Gen1 tables, which are reported on the test split this release does not carry (Sec. 7). Resampling the same 406 sequences and re-ranking within each replicate, that validation-split ordering is reproduced in 80.5% of replicates on the moving subset and 42.5% over every labeled box, the population the released evaluation scores. Its tightest adjacent pair is separated by 0.050 points against a paired standard error of 0.53 and exchanges places in a fraction 0.48 of replicates. Two readings are consistent with the same evidence: that the correction changes no ranking on this split, and that the ranking is unstable under sequence resampling. The measured sampling uncertainty separates them in favor of the second.

Normalized image speed and timing resolution. The natural reading of main Tab. 2 is that a benchmark of faster objects would resolve the interval. What an IoU-thresholded metric registers is displacement relative to box size, so the scale-free quantity is  $r = |v|/\sqrt{wh}$ , box-widths of motion per second, which is invariant to sensor resolution and therefore comparable across benchmarks. A single fixed summary of  $r$ , such as its median, does not explain the observed resolution across strata. Scaling  $\delta$  by each stratum's median  $r$  increases the dispersion of the 5%-loss timescale from 0.20 to 0.60, and no fixed quantile improves on the unscaled result.

mAP thresholds each box rather than averaging over them. Under the single-threshold caricature, in which a box loses its match once displaced past  $d^*$  box-widths and is untouched before, the fraction of boxes lost at displacement  $\delta$  is  $P(r > d^*/\delta)$ , so the displacement at which a loss level  $p$  is reached is  $D(p) = d^*/Q_{1-p}(r)$ . The scale-setter is the quantile matched to the loss level being read off, rather than the median or any fixed quantile. Scaled that way the strata do collapse, to a coefficient of variation of 0.042 at the 5% level over three strata and 5 checkpoints, with  $d^* = 0.0628 \pm 0.0029$  box-widths. That constant also follows from the matching geometry. Displacing a square box by a fraction  $d$  of its

width gives  $\text{IoU} = (1 - d)/(1 + d)$ , so  $d^*$  is the displacement that breaks a match at  $\text{IoU} = 0.882$ , inside the 0.50:0.05:0.95 range this mAP averages over and near its top, where the first few percent of loss arise. The model degrades as the caricature should, to 0.10 at 10% loss and 0.28 at 25%. The  $\geq 50$  px/s stratum is the outlier at every loss level, with  $d^*$  approximately  $1.8\times$  larger than the common value. Its lower mAP(0) may leave less high-IoU headroom for displacement, but no downstream claim depends on this interpretation.

These results show that timing resolution is governed by the upper tail of  $r$ . Stratification increases the relevant quantile but also increases the standard error of  $\Delta\text{mAP}$  as  $1/\sqrt{n}$ . The 25–50 px/s and  $\geq 50$  px/s strata yield net resolution factors of  $0.42\times$  and  $0.27\times$ , respectively. Neither resolves the interval better than the full moving-box population.

On the DSEC-Det labels, associated by the `track_id` the release ships rather than by IoU linking,  $r$  has median  $1.31\text{s}^{-1}$  against Gen1's  $0.095\text{s}^{-1}$ . On the quantity that actually governs resolution the gap is smaller:  $Q_{95}(r) = 5.59$ , compared with 0.70 on Gen1, a  $7.97\times$  difference. This label-composition contrast shows that DSEC-Det has a substantially heavier upper tail of normalized image speed than Gen1. It supports the narrower conclusion that Gen1's timing insensitivity should not be generalized to event-based detection benchmarks with different motion and box-size distributions.

### 13. Recurrent support under the released streaming evaluation

The previous section established that the carried state of the SSM release is inert, so under its own streaming evaluation the recurrent history behind a detection is bounded by the current chunk of 21 windows. That bound is not constant across detections: a frame at the first position of a chunk carries one window, and a frame at the last carries 21. The reported Gen1 number averages over all 21 positions, and neither the metric, the protocol, nor the published table names the variable.

The position is recoverable without running anything. The release derives its chunk starts from the index files it ships: the first chunk begins at

$$\begin{aligned} &\text{max}(\text{objframe\_idx\_2\_repr\_idx}[0] \\ &\quad - \text{sequence\_length} + 1, 0) \end{aligned}$$

and later chunks step by `sequence_length`, so the position of every labeled frame follows from those files alone. Reconstructing it over the 20 296 labeled frames

reproduces, position by position, the positions the instrumented dumper records at run time.

Comparing the frames at positions 0–3 with those at positions 16–20 confounds support with scene content, since these are different frames. The three RVT checkpoints control for that: they see the identical frames in the identical order, but their ConvLSTM state is carried across chunks, so position in the chunk is not a support variable for them and any difference they show is scene content. Differencing removes it. Tab. 4 scores every labeled box, the population the released numbers use, and bootstraps over the 406 sequences.

| checkpoint | short<br>(1–4 win) | full<br>(17–21 win) | raw $\Delta\text{mAP}$<br>(pt) | net of RVT control<br>(pt) |
|------------|--------------------|---------------------|--------------------------------|----------------------------|
| rvt-t      | 38.180             | 38.194              | 0.01                           | $-0.67 \pm 0.43$           |
| rvt-s      | 38.371             | 39.167              | 0.80                           | $0.51 \pm 0.48$            |
| rvt-b      | 41.561             | 42.126              | 0.56                           | $0.16 \pm 0.55$            |
| S5-S       | 31.578             | 36.410              | 4.83                           | $4.37 \pm 0.61$            |
| S5-B       | 34.445             | 40.226              | 5.78                           | $5.32 \pm 0.53$            |

Table 4. mAP over every labeled box, by the amount of recurrent history the released streaming evaluation gives the frame. Raw  $\Delta\text{mAP}$  is the full column minus the short one, the quantity main Fig. 2 plots, so positive means longer history scored higher. The last column subtracts the placebo trend from it, using the mean of the other two RVT rows for an RVT row and the mean of all three for an SSM row. The RVT rows are the placebos: their state crosses chunk boundaries, so position carries no support for them. Net of that control the SSM rows gain 4.37 and 5.32 points ( $z = 7.2$  and  $10.0$ ), against at most 0.67 points in magnitude ( $z = 1.56$ ) for any placebo. The short and full columns carry three decimals, so differencing them reproduces the raw column exactly. Cluster bootstrap over 406 sequences.

The RVT placebos isolate chunk-position effects under the parallel-trends assumption. The RVT checkpoints span the same 38.180–41.561 mAP range and are scored on the same frames, and their rows show no comparable shift, which argues against frame difficulty at early chunk positions as the source (Fig. 4).

A permutation null tests whether the reconstructed correspondence itself can account for the effect. Chunk position is reshuffled among the frames of each sequence, which preserves the scene, the sequence lengths, the label density and the sizes of both blocks, and destroys only the correspondence between a frame and the history the protocol gave it. Over 200 reshuffles the difference-in-differences is centered on zero with a standard deviation of 0.55 points, and the observed values lie 9.8 and 7.6 standard deviations outside it ( $p = 0.005$ , the smallest value 200 reshuffles can return), while every placebo stays within 1.52. The effect disappears when the correspondence between a frame and its reconstructed history is permuted.

A second instrument uses neither the ground truth nor the evaluator. Mean detection confidence is a label-

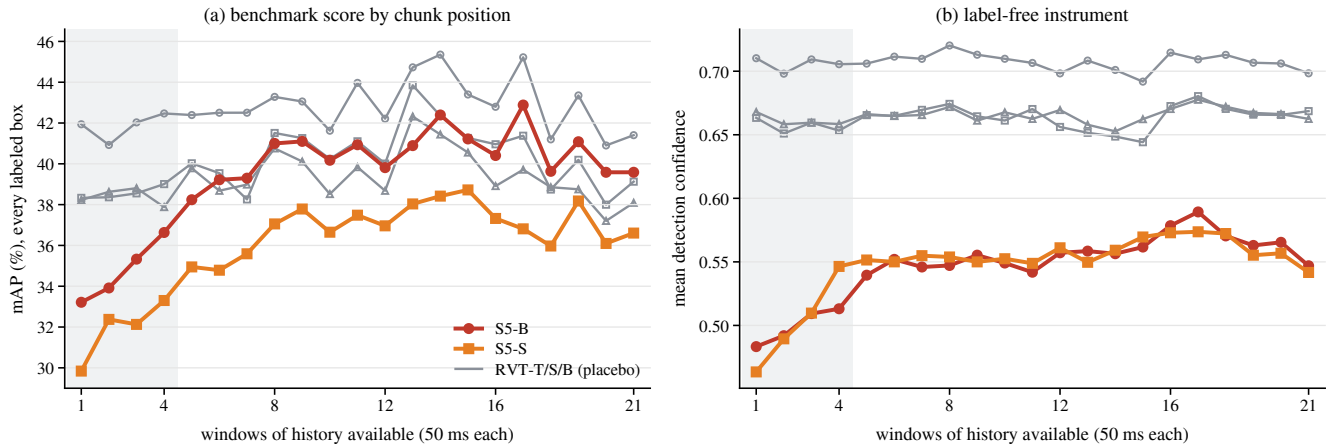


Figure 4. The chunk-position contrast at each of the 21 positions. (a) mAP over every labeled box, the main-paper Fig. 2 at full width. (b) Mean detection confidence at the same positions, which uses neither the ground truth nor the evaluator. Both panels separate the two SSM checkpoints from the three RVT placebos in the same direction, so the contrast is present upstream of the matching rule. Shaded: the one-to-four-window block.

free statistic, and it is flat in chunk position when the available support does not vary with that position. For S5-S it rises monotonically from 0.463 at one window to 0.574 at seventeen, 23.8%, while rvt-t stays flat across the same positions. The shift therefore appears upstream of the matching rule.

The comparison can also be made against the reported figure itself. Under the released evaluation, the pooled S5-B and S5-S scores lie 1.75 and 1.33 points below the scores of their respective full-support frames.

### 13.1. The same frame under both chunk positions

Every estimate above compares different frames and therefore rests on a parallel-trends assumption: absent the support difference, frames that fall early and late in a chunk are assumed to differ only by a model-independent offset. The placebos make that assumption testable, but they do not remove it.

The release’s chunk starts follow from the index files, so moving them is a one-line change to the protocol and not to the model. Shifting every start 5 windows later sends the frame the release gave position  $p$  to position  $(p - 5) \bmod 21$ , which carries the starved block, positions 0–3, onto 16–19. Those frames are then scored twice: once with one to four windows of recurrent history, as the release scores them, and once with seventeen to twenty, under the same weights, the same ground truth and the same evaluator. 3574 labeled frames satisfy both, and the comparison is within model and within frame. Shifting earlier does not work: the release’s own first start is already clamped at zero for most sequences, so most frames would keep the position they had.

Both paired estimates move in the same direction as

| paired quantity, 3574 frames          | S5-B                | S5-S                |
|---------------------------------------|---------------------|---------------------|
| $\Delta$ mAP, velocity-evaluable (pt) | $5.23 \pm 0.49$     | $4.81 \pm 0.52$     |
| $\Delta$ mAP, every labeled box (pt)  | $4.20 \pm 0.36$     | $4.10 \pm 0.32$     |
| $\Delta$ mean confidence              | $0.0579 \pm 0.0048$ | $0.0509 \pm 0.0035$ |
| $\Delta$ mean confidence, matched     | $0.0409 \pm 0.0040$ | $0.0519 \pm 0.0043$ |
| $\Delta$ detections per frame         | $-3.49 \pm 0.25$    | $-5.09 \pm 0.26$    |

Table 5. The identical frames scored under one to four windows of recurrent history and under seventeen to twenty, the sign taken as the long arm minus the short one. Neither a control model nor a parallel-trends assumption enters this paired comparison. Cluster bootstrap over the same 406 sequences with 300 replicates. The mAP row has  $z = 10.7$  and  $9.3$ , mean confidence 12.0 and 14.6. Matched confidence is over detections paired to a label of the same class at IoU  $\geq 0.5$ .

the chunk-position contrast. Their magnitudes are not compared directly because the velocity-evaluable and all-label populations differ. The remaining two rows of Tab. 5 show what moves. Confidence rises on the detections that match a label, by 0.0409 and 0.0519, and total output falls by 3.49 and 5.09 boxes per frame, so both terms of precision move the same way: the starved arm emits more boxes and scores the real ones lower. Neither channel is available to a reader of the pooled number, which averages over the chunk position that sets them.

## 14. The recurrent influence tail

Main Sec. 3.3 reports that the window-ablation-sensitivity centroid tracks the horizon rather than converging. Two horizons alone cannot distinguish a slowly converging support from a divergent one, so the tail is characterized directly.

Let  $I(L)$  be the summed sensitivity at lag  $L$  win-

dows. A weighted least-squares fit of  $\log I$  on  $\log L$  over lags 1–9 gives  $I(L) \propto L^{-a}$  with  $a = 0.628$ , and the convergent alternative  $I(L) \propto e^{-L/\lambda}$  was fitted on the same lags. Neither was fitted on lags 10–19, which are held out. On the held-out lags, the power law gives an RMS standardized residual of 2.5, compared with 23.8 for the exponential, a  $9.4\times$  difference. Fitting all lags 1–19 gives  $a = 0.645 \pm 0.005$ , with a bootstrap interval of  $[0.636, 0.655]$  that no replicate carried to  $a \geq 1$ . The standard error and that interval come from a parametric resample of the per-lag means and so describe the precision of the fit, not independent observations: all lags are computed from the same 576 samples. These lags are not treated as independent observations.

For a support whose first moment has converged, the centroid ratio approaches 1 as the horizon increases, and a particular finite doubling need not give exactly 1. The measured ratio is 1.864  $[1.851, 1.876]$ , an interval excluding 1, and follows from two measured centroids without a fit, so over the horizons measured here the centroid is still moving with the horizon rather than settling. Two finite horizons cannot separate a slowly converging tail from a divergent one, and the exponent is fitted over the observed lag range. The measured horizons do not determine a converged full-history centroid. Conditionally, if the observed  $I(L) \propto L^{-a}$  with  $a < 1$  persisted asymptotically, then both  $\sum_L I(L)$  and  $\sum_L L I(L)$  would diverge and the centroid over a horizon of  $H$  windows of width  $w$  would be  $-wH(1-a)/(2-a)$ , linear in  $H$ , and the first moment would then not exist. Either way 77.3% of the summed window-ablation sensitivity already lies outside the current window, which is what the identifiability claim rests on.

A fill artifact could in principle produce persistent long-lag sensitivity, so the control defined in Sec. 2 applies here. The zero and swap fills decay by factors of 10.3 and 2.2 over the same lags, while substantial long-lag sensitivity remains detectable under both fills.